

Project Title

Serverless Student Data Processing System Using AWS

Introduction

A serverless application using AWS Lambda, API Gateway, DynamoDB, and CloudWatch to process and store student data.

Objectives

- Learn serverless computing
- Integrate API Gateway with Lambda
- Store data in DynamoDB
- Monitor using CloudWatch

Technology Used

AWS Lambda
Amazon API Gateway
Amazon DynamoDB
Amazon CloudWatch
IAM
Python 3.12
Postman

Methodology

1. Create DynamoDB table
2. Create Lambda
3. Configure IAM
4. Write Python code
5. Configure API Gateway
6. Test with Postman
7. Verify in DynamoDB
8. Monitor in CloudWatch

Architecture

Client/Postman → API Gateway → Lambda → DynamoDB → CloudWatch

Output / Result

Successfully inserted records into DynamoDB and triggered Lambda through API Gateway.

Conclusion

Implemented a complete serverless AWS solution without managing servers.